AJAY A

Electronics and Communication Engineer
ajayjesus66@gmail.com | LinkedIn | 9342371647 | Thoothukudi,Tamilnadu

Career Objective

Electronics and Communication Engineering student with strong fundamentals in analog and digital electronics and circuit design, backed by skills in C, Verilog HDL, and MATLAB. Aspiring to build a career in core electronics, embedded systems, or hardware design by applying technical expertise, analytical thinking, and problem-solving skills to real-world engineering challenges.

B.E Electronics and Communication Engineering
University VOC College of Engineering,Thoothukudi
2023-2027|CGPA: 7.47(till 6th semester)

HSC
St.Thomas Mat Hr Sec School,Thoothukudi
2023|Percentage: 91.3%

Skill set

- **Programming:** C, MATLAB, Verilog HDL
- **Circuit Simulation Design:** Pspice, KiCad
- **Tools:** Arduino IDE,Wokwi,Tinkercad,KiCad,ModelSim,Quartus II

Academic projects

Smart India Hackathon 2026-Predictive Solar Energy Management & Load Balancing[Ongoing]

- Developed a predictive solar–battery–grid energy management system using weather forecasting and PuLP-based optimization for cost-aware power dispatch.
- Implemented battery management with SOC limits, charge/discharge constraints, efficiency losses, and degradation modeling.

Over-the-Air (OTA) Firmware Update System

- Developed an OTA firmware update system for secure remote software updates over WiFi.
- Improved device maintenance by reducing manual intervention and enhancing system reliability.

IoT-Based Smart Waste Management System

- Developed an IoT-enabled smart waste management system using ESP8266, an ultrasonic sensor, and a servo motor.
- Integrated the Blynk platform for remote monitoring, demonstrating skills in IoT, embedded systems, and sensor interfacing.

Fingerprint-Based Door Access Control System

- Developed a secure embedded door access control system using fingerprint authentication for authorized user verification and entry.
- Gained hands-on experience in fingerprint sensor interfacing, microcontroller programming, embedded systems, and hardware integration.

Internship

PCB Design and Testing Intern Quantanics TechServ Pvt.Ltd.,Madurai 15/06/26-15/07/26

- Successfully completed a one-month internship focused on PCB design, schematic capture, layout, routing, and testing using EDA tools..

Certificates

- Attended a workshop on "Embedded 8051 Interfacing"
- Participated in Internal SIH2026

Areas of Interest

Automobile Industry | Embedded systems | IoT